

25/6/25

## SQT Assignment - 1

1) What is Software Testing? [Explain about SDLC and STLC.] x

Ans. → Software Testing is an important process in the Software Development Life Cycle (SDLC). It involves ~~verify~~ verifying & validating that a Software Application is free of bugs, meets the technical requirements set by its Design and Development, and satisfies user requirements efficiently and effectively.

→ Software Testing is mainly divided into two parts, which is used in the Software Development Process:-

① Verification - This step involves checking if the software is doing what is supposed to do.

② Validation - This step verifies that the software actually meets the customer's needs and requirements.

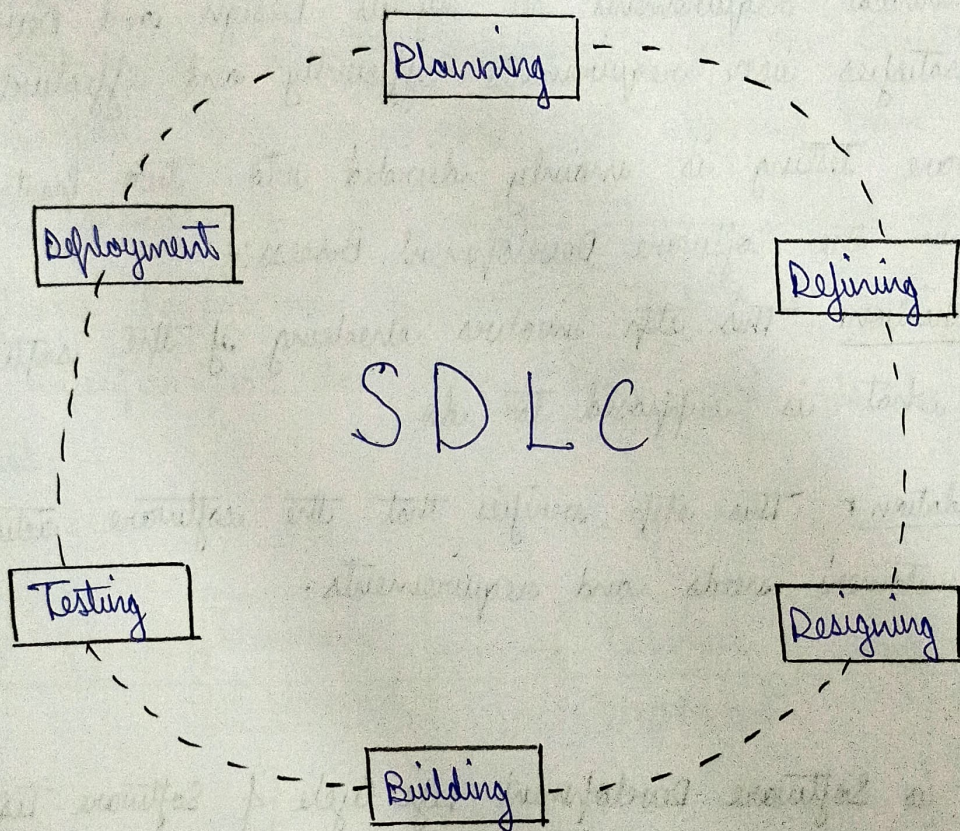
2) What is Software Development Life Cycle [ & Software Testing Life Cycle.] x

Ans. → Software Development Life Cycle (SDLC) is a structured process that is used to design, develop, and test good-quality software. SDLC is a methodology that defines the entire procedure of software development step-by-step.

→ The goal of SDLC is to deliver high-quality maintainable software that meets the user's requirements. SDLC in software engineering outlines the plan for each stage so that



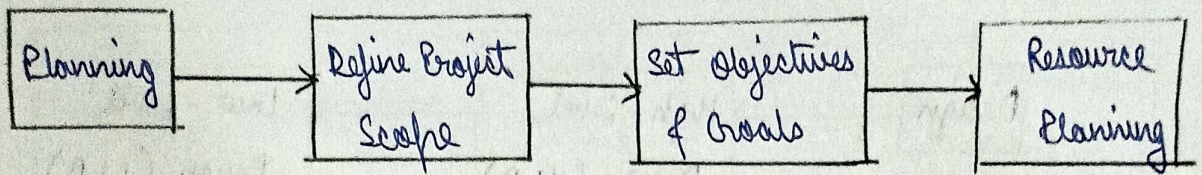
each stage of the software development model can perform its task efficiently to deliver the software at a low cost within a given time frame that meets users' requirements.



### Stage - 1 :- Planning and Requirements Analysis

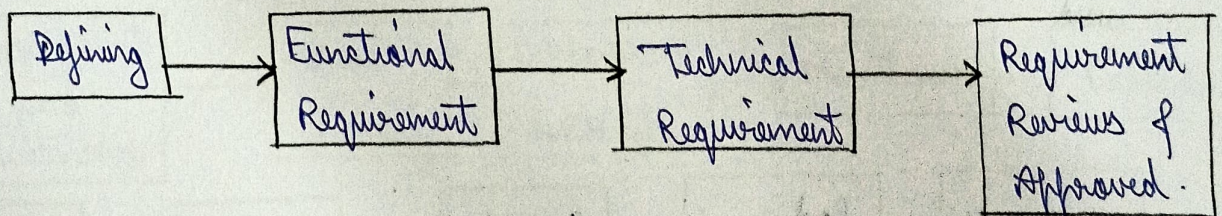
→ Planning is a crucial step in everything, just as in software development. In the same stage, requirement analysis is also performed by the developers of the organization. This is obtained from customer inputs, & sales department / market surveys.





### Stage 2:- Refining Requirements

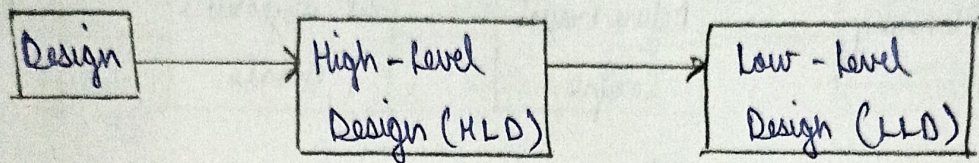
→ In this stage, all the requirements for the target software are specified. These requirements get approval from customers, market analysts, & stakeholders. This is fulfilled using SRS (Software Requirement Specification). This is a sort of document that specifies all the things that needs to be defined & created.



### Stage 3:- Designing Architecture

→ SRS is a reference for software designers to come up with the best architecture for the software. Hence, with the requirements defined in SRS, multiple designs for the product architecture are present in the Design Document Specification (DDS)

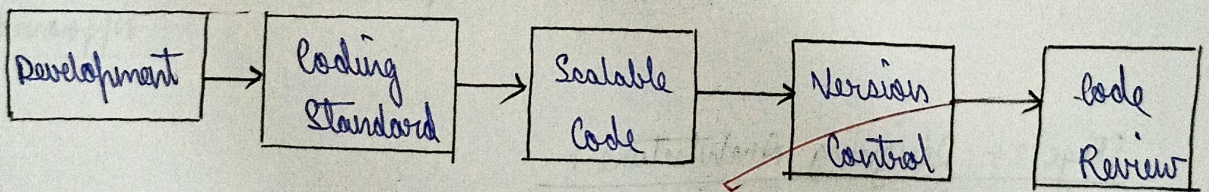




#### Stage 4:- Developing Product

→ At this stage, the fundamental development of the product starts. For this, developers use a specific programming code as per the design in the DDC. Hence, it is important for the coders to follow the protocols set by the association.

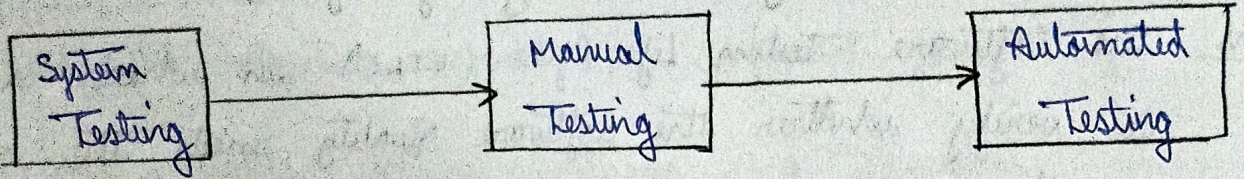
Programming Tools like compilers, interpreters, debuggers, etc of programming languages like C/C++, Python & Java are used.



#### Stage 5 :- Product Testing & Integration

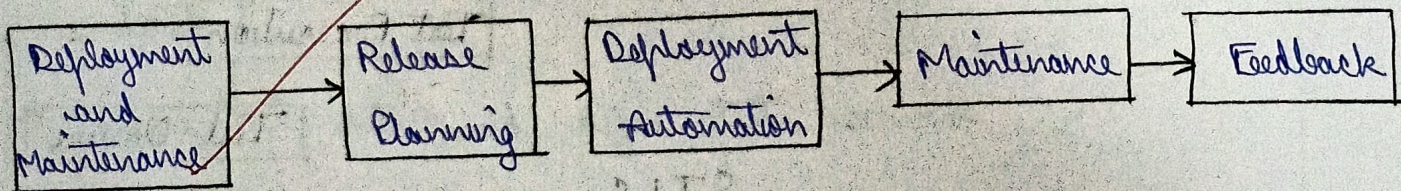
→ Software documentation is an essential part of the software development life cycle. A well-written document acts as a tool & means to information repository necessary to know about the software process, functions & maintenance. Documentation also provides information about how to use the project.





### Stage 6:- Deployment and Maintenance of Products:-

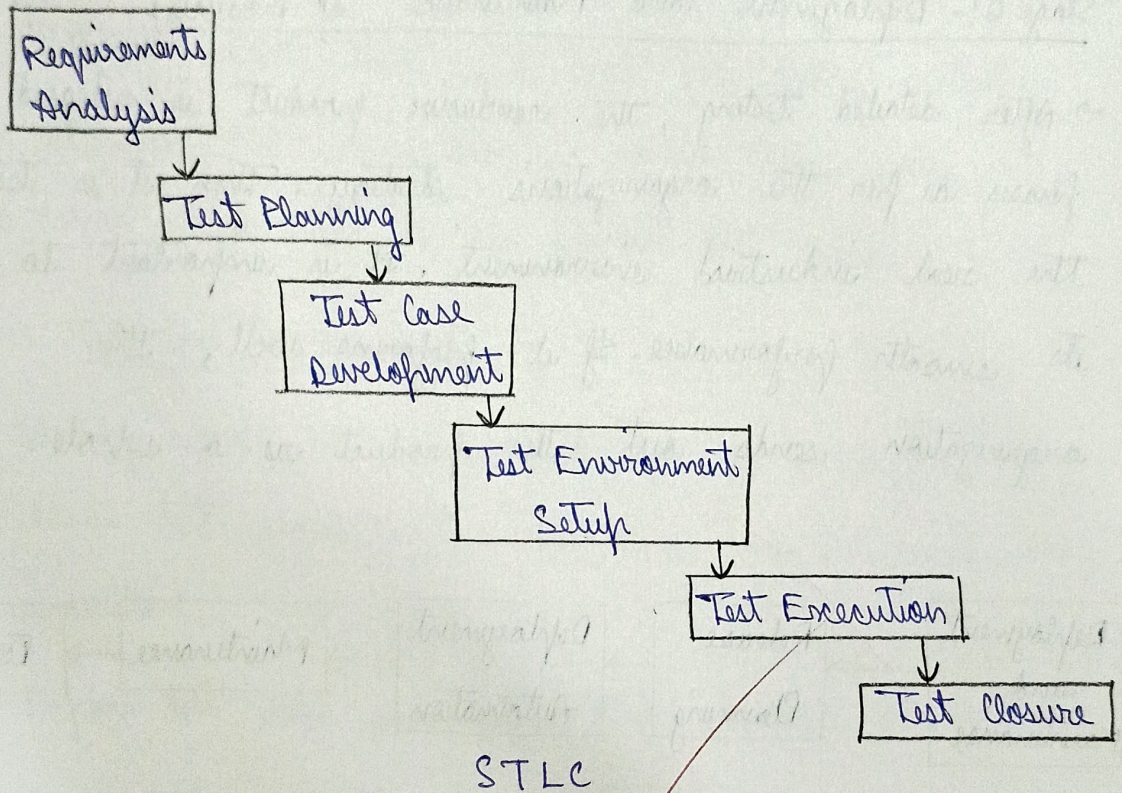
→ After detailed testing, the conclusive product is released in phases as per the organization's strategy. Then it is tested in the real industrial environment. It is important to ensure its smooth performance. If it performs well, the organization sends out the product as a whole.





3) Explain about Software Testing Life Cycle (STLC).

Ans The Software Testing Life Cycle (STLC) is a process to verify whether the Software Quality meets the expectations or not.



#### (1) Requirement Analysis :-

- Understand what needs to be tested by reviewing the Software Requirements Document (SRD).
- Identify any missing, incomplete, or ambiguous requirements.
- Interview stakeholders to clarify doubts.
- Analyze potential risks or issues that may impact testing.
- Create a Requirement Traceability Matrix (RTM) to map requirements to test cases.



### (ii) Test Planning

- Define the objectives, scope and approach for testing.
- Choose suitable testing methods and techniques.
- Identify necessary test environments, tools and resources.
- Estimate effort, time and cost for testing activities.
- Determine test deliverables & assign team roles.
- Review and approve the overall test plan.

### (iii) Test Case Development

- Identify which test cases need to be developed.
- Write test cases that are clear, concise and easy to understand.
- Prepare test data and define test scenarios.
- Specify the expected results for each test case.
- Review and validate test cases for accuracy.
- Update the RTM to ensure all requirements are covered.

### (iv) Test environment setup

- Set up the hardware, software and network needed for testing.
- Ensure all tools, configurations, and test scripts are ready.



→ This setup can be done independently of the development team.

→ Prepare the environment to match real - world usage.

#### (v) Test Execution

→ Execute the test cases prepared in the earlier phase

→ Log any defects found, including severity and priority

→ Analyze test results to assess performance & identify issues

→ Retest after defects are fixed to confirm resolution

→ Report the test execution progress & findings.

#### (vi) Test Closure

→ Create a test summary report that includes number of tests run, defects found, and pass / fail rate.

→ Ensure all defects are tracked & resolved

→ Clean up the test environment & remove test data.

→ Prepare a test closure report detailing activities, schedules, and results.

→ Share knowledge with the team & gather feedback for future improvements.

~~Review~~